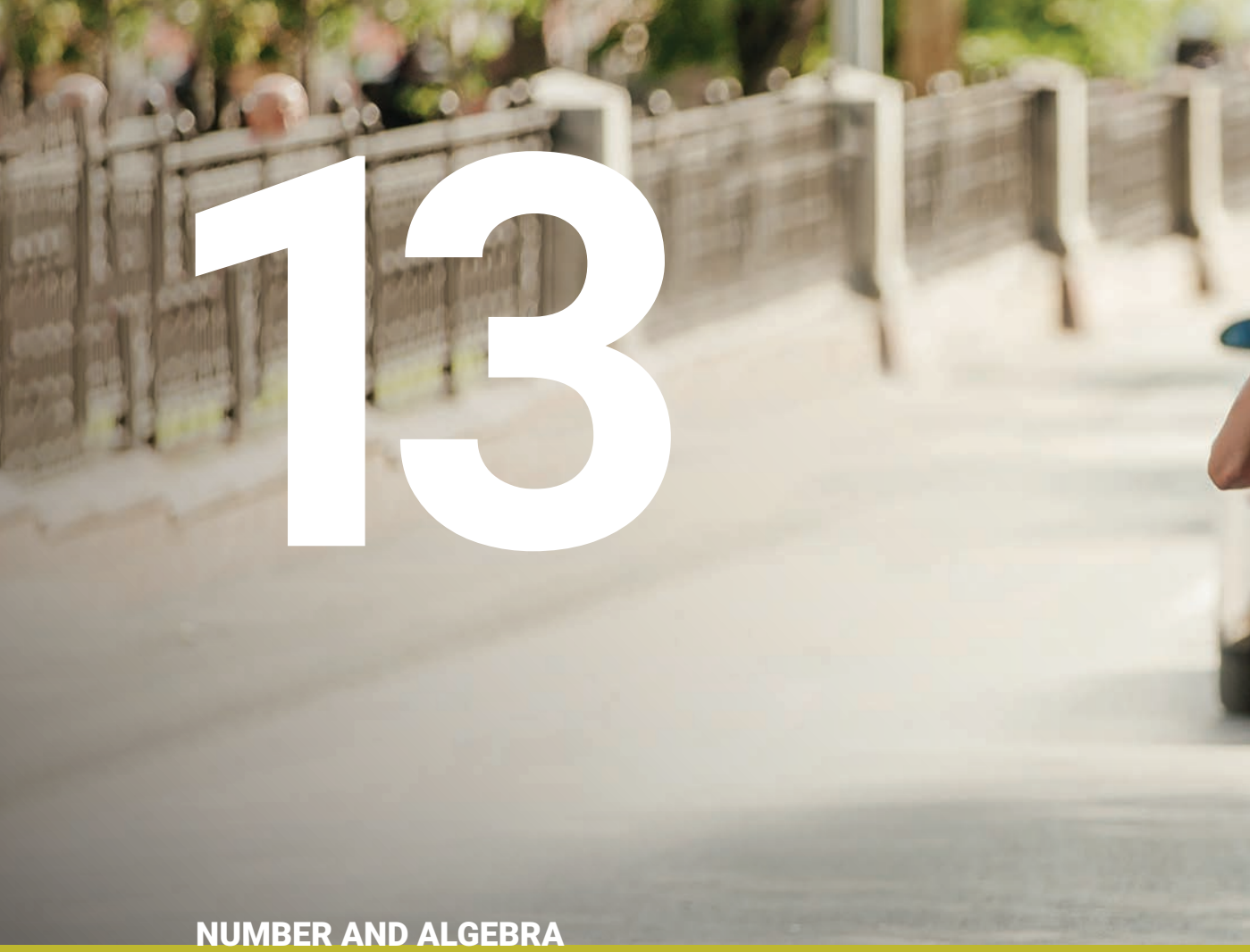


13

NUMBER AND ALGEBRA

FUNCTIONS

OPTIONAL PATH TOPIC
FOR HSC MATHEMATICS ADVANCED

Functions are used in many areas of mathematics and science and can be used to describe the relationship between 2 quantities such as the distance run by an athlete and the time taken. In business, computing, economics and engineering, finding the right model or function is important for solving problems. The German mathematician, Peter Dirichlet, was the first person to use the word *function* in 1837.



sportpoint/Shutterstock.com

Chapter outline

		Working mathematically				
13.01	Functions*	U	F		R	C
13.02	Function notation*	U	F		R	C
13.03	Graphing translations of functions*	U	F		R	
13.04	Linear inequalities on the number plane*	U	F		R	C

***PATH**

Wordbank

domain The set of permissible x values for the function $y = f(x)$

function A rule or relationship between 2 variables where, for each value of the independent variable (input value), x, there is only one value of the dependent variable (output value), y

function notation A way of writing a function, using the form $y = f(x)$

range The set of permissible y values for the function $y = f(x)$

vertical line test A test applied to a graph to determine whether it is the graph of a function

Quiz
Wordbank 13

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Quiz (1):
- Wordbank 13

Worksheets (4):
13.02 Function notation • Algebra 5
13.03 Graphing translations of functions
13.04 A page of lines and regions

Nelson MindTap

To access resources above, visit
cengage.com.au/nelsonmindtap

In this chapter you will:

- (PATH) understand the definition of a function and use function notation
- (PATH) use the vertical line test on a graph to test whether it represents a function
- (PATH) find the domain and range of a function
- (PATH) sketch $y = f(x) + c$ and $y = f(x - a)$ given the graph of $y = f(x)$
- (PATH) graph linear inequalities on a number plane, such as $2x - 3y > 5$

13.01 Functions

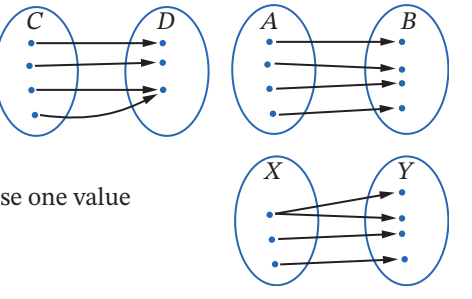
PATH

A **function** is a relationship between 2 variables, usually described by an equation such as $y = 3x^2 - 5x + 1$. In this example, the value of y depends on the value of x, so y is called the **dependent variable** and x is called the **independent variable**.

Functions

- A **function** is a relationship between 2 variables where, for each value of the independent variable, there exists **only one** value of the dependent variable.
- If x is the independent variable and y is the dependent variable, then for each x value, there is **only one** y value.

A function can also be visualised as a relationship between 2 sets of values where every value of one set associates with (or maps to) **exactly one** value of the second set. Two examples of functions are shown here:



This next diagram does *not* represent a function because one value of set X associates with (or maps to) **2** values of set Y.

Example 1

Do the points (3, 5), (4, 3), (4, −2) represent a function?

Solution

When $x = 4$, $y = 3$ and $y = -2$.

So, one x value gives 2 y values.

∴ The points do not represent a function.

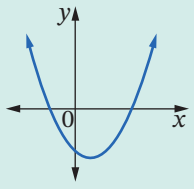
Vertical line test

- Any vertical line drawn on the graph of a function will cut the graph **only once.**
- If any vertical line can cut a graph in more than one place, then the graph is **not** a function.
- This is a visual representation of the definition: 'for each x value, there is **only one** y value'.

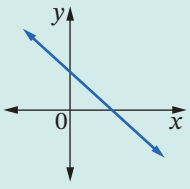
Example 2

Use the vertical line test to determine whether each graph represents a function.

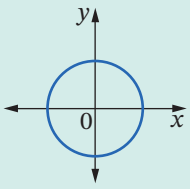
a



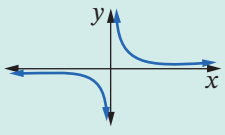
b



c

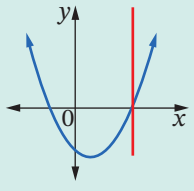


d

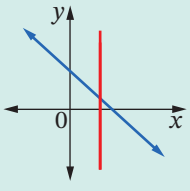


Solution

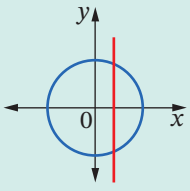
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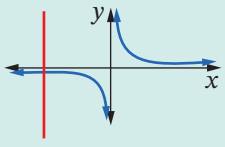
b



c



d



a Any vertical line will only cut the parabola once.

∴ The graph represents a function.

For every x value, there is only one y value.

b Any vertical line will only cut the line once.

∴ The graph represents a function.

c A vertical line can cut the circle twice.

∴ The graph does not represent a function.

For this x value, there are 2 y values.

d Any vertical line will only cut the hyperbola once.

∴ The graph represents a function.

PATH

EXERCISE 13.01 ANSWERS ON P. 680

Functions U F R C

EXAMPLE 1

1 Determine whether each set of points represents a function. R C

a (2, 4), (−2, 7), (5, 2)

b (−1, 7), (−1, 3), (4, 6)

c (−6, 3), (5, 1), (4, −5), (−3, −1)

d (2, 3), (−5, −3), (4, 4), (8, −3)

e (5, 8), (−1, −1), (3, 8), (4, 0)

f (7, 2), (−6, 0), (0, 3), (−3, 2)

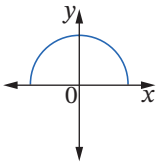
g (3, −1), (3, 5), (3, −6), (3, 0)

h (0, 2), (7, 2), (−4, 2), (9, 2), (5, 2)

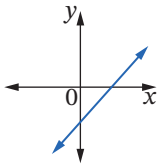
EXAMPLE 2

2 Test whether each graph represents a function. R C

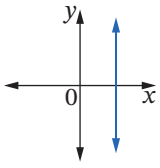
a



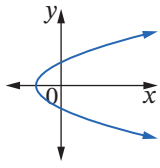
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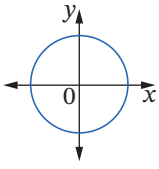
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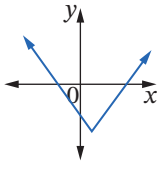
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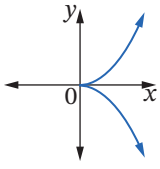
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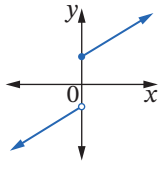
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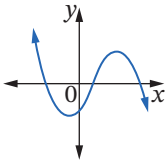
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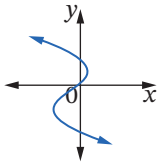
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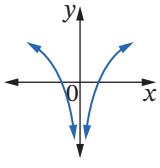
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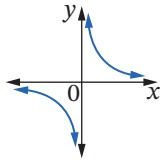
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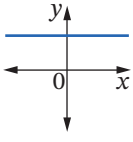
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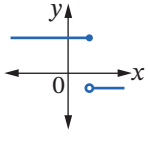
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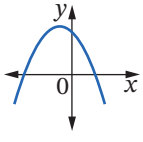
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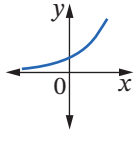
n



o



p



3 **a** Graph each linear equation and find its gradient. R C

i $y = 3x - 2$

ii $x + 2y = 3$

iii $y = 4$

iv $x = -2$

b Which of the lines drawn in part **a** represent a function?

c For what values of the gradient is a linear equation:

i always a function?

ii never a function?

Foundation Standard Complex

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Function notation

13.02

For the function $y = 3x - 1$, y depends on x.

We say that 'y is a function of x'. This is written as $y = f(x)$, which is read as 'y is f of x' and the 'f' stands for function.

So, for $y = 3x - 1$, we may also write $f(x) = 3x - 1$.

Note that other variables can be used in functions. For example, if the rule $h = 12t - 5t^2$ represents the height, h metres, of a ball after time t seconds, then h is a function of t and we can write $h(t) = 12t - 5t^2$.

PATH



Worksheet
Function notation

13.02

The value of a function

If $y = f(x)$, then $f(c)$ is the value of the function when $x = c$ (or the y value when $x = c$).

Example 3

If $y = f(x) = x^2 - 2x$, find:

a $f(3)$ **b** $f(-4)$ **c** $f(2a)$

Solution

a $f(x) = x^2 - 2x$

$\therefore f(3) = 3^2 - 2 \times 3$

$= 9 - 6$

$= 3$

b $f(-4) = (-4)^2 - 2 \times (-4)$

$= 16 + 8$

$= 24$

c $f(2a) = (2a)^2 - 2 \times 2a$

$= 4a^2 - 4a$

Example 4

For the function $g(h) = 3h + 5$, find:

a $g(-2)$ **b** $g(3) + g(-1)$

c $g(k + 4)$ **d** the value of h for which $g(h) = 20$

Solution

a $g(-2) = 3(-2) + 5$

$= -6 + 5$

$= -1$

b $g(3) = 3(3) + 5$

$= 14$

$g(-1) = 3(-1) + 5$

$= 2$

$\therefore g(3) + g(-1) = 14 + 2$

$= 16$

c $g(k + 4) = 3(k + 4) + 5$

$= 3k + 12 + 5$

$= 3k + 17$

d $g(h) = 3h + 5 = 20$

$3h = 20 - 5$

$= 15$

$h = 5$

PATH

The domain and range of a function

For some functions $y = f(x)$, there may be restrictions on the x values and y values. The set of permissible x values is called the **domain**, while the set of permissible y values is called the **range**.

Example 5

Draw the graph of each function and find its domain and range.

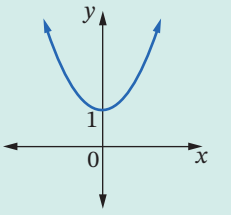
a $f(x) = x^2 + 1$ **b** $y = \frac{3}{x}$ **c** $y = 3^{-x}$

Solution

a The graph of $y = f(x) = x^2 + 1$ is a parabola with a vertex at (0, 1).

Note from the equation and graph that there is no restriction on the x values, so the **domain** of the function is 'all x'.

Note from the equation and graph that the y values are all greater than or equal to 1. So the **range** of the function is $y \geq 1$.



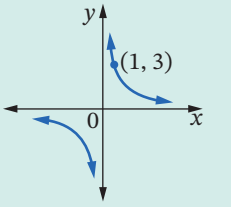
b The graph of $y = \frac{3}{x}$ is the hyperbola as shown.

The permissible x values are all values of x, except 0, which is written as $x \neq 0$.

The **domain** of the function is $x \neq 0$.

The permissible y values are all y values except 0.

The **range** of the function is $y \neq 0$.

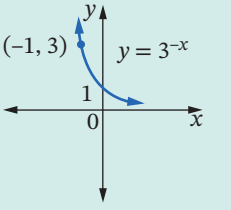


c The graph of $y = 3^{-x}$ is the exponential curve as shown.

The **domain** is 'all x'.

The permissible y values are all positive y values.

The **range** is $y > 0$.



EXERCISE 13.02 ANSWERS ON P. 680

EXAMPLE 3

Function notation U F R C

1 For each function below, evaluate:

	i $f(3)$	ii $f(-1)$	iii $f(0)$
a $f(x) = 2x$		**b** $f(x) = 1 - x$	**c** $f(x) = x^2 + 5x$
d $f(x) = \frac{1}{2x+1}$		**e** $f(x) = \sqrt{2x+3}$	**f** $f(x) = \frac{2x+1}{2x-1}$
g $f(x) = 1 - \frac{2}{1-2x}$		**h** $f(x) = x^2 - 4x + 1$	**i** $f(x) = (x+5)^2 - 4$

Worksheet
Algebra 5

Foundation Standard Complex

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2 If $f(x) = 3x - x^2$, find: R C

a $f(3)$ **b** $f(-4)$ **c** $f(3k)$

3 If $f(x) = 6x - 2$, find: R C

a $f(3)$ **b** $f(-2) + f(1)$ **c** $f(d - 3)$

d $f(d + 1) - f(d - 1)$

e the value of x for which $f(x) = 15$

4 If $h(x) = \frac{x+2}{x-1}$, find: R C

a $h(2)$ **b** $h(-1)$ **c** $h(0)$ **d** x, if $h(x) = 3$

5 If $g(t) = \frac{t}{2} + t^2$, find: R C

a $g(2)$ **b** $g(0)$ **c** $g(-4)$

d the values of t for which $g(t) = 3$

6 Given $p(y) = 11 - 2y$, find: R C

a **i** $p(0)$ **ii** $p(-4)$ **iii** $p(0.5)$

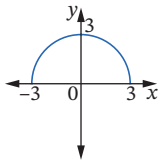
b **i** $p(y + 1)$ **ii** $p(y + 1) - p(y)$

c y given that $p(y) = -21$.

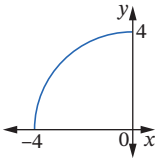
7 For each graph of a function, find: R C

i the domain **ii** the range

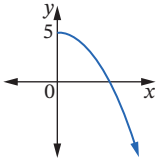
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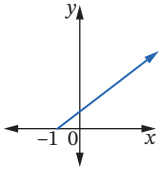
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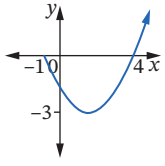
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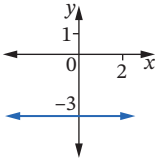
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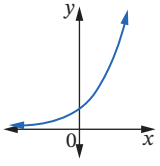
e



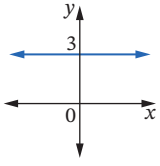
f



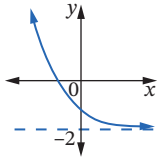
g



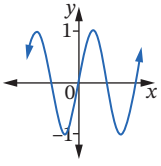
h



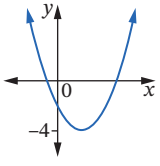
i



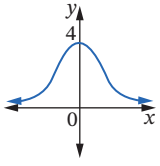
j



k



l



PATH

8 Graph each function and write its domain and range. R C

a $f(x) = 3x + 1$ **b** $y = 1 - x^2$ **c** $f(x) = 2^x$

d $y = \frac{2}{x}$ **e** $y = 1 - 2x$ **f** $f(x) = x^2 - 2x - 15$

g $f(x) = x^3$ **h** $f(x) = 4^x + 3$ **i** $y = (x + 3)^2 + 2$

9 $g(m) = m^2 + 2$ and $h(m) = 2m + 5$. R C

a Find:

i $g(3)$ **ii** $h(-5)$ **iii** $g(-1) + h(-1)$

b Show that $h(w^2) - 2g(w) = 1$.

c Solve $g(m) = h(m)$.

10 $k(x) = \sqrt{2x+1}$. R C

a Find $k(4)$. **b** Solve $k(x) = 5$.

c Explain why there is no value for $k(-1)$.

d What is the smallest value for x such that $k(x)$ exists?

11 If $h(t) = 3t^2 - 2t + 3$, find: R C

a $h(-1)$ **b** $h(2)$ **c** $h(4) - h(0)$

d $h(k^2)$ **e** $h(-k)$ **f** $h(k) - h(-k)$

12 $H(x) = 2^x + 2^{-x}$. R C

a Find the values of $H(3)$, $H(-2)$ and $H(a + 1)$.

b Show that $H(x) = H(-x)$.

13 $F(x) = x - 3 + \frac{1}{x}$. R C

a Evaluate:

i $F(3)$ **ii** $F(-3)$ **iii** $F\left(\frac{1}{3}\right)$

b Does $F(a) = F(-a)$ for all possible values of a?

c Does $F(a) = F\left(\frac{1}{a}\right)$ for all possible values of a?

13.03 Graphing translations of functions

PATH



Worksheet
Graphing translations of functions

Vertical and horizontal translations of $y = f(x)$

- The graph of $y = f(x) + c$ is the graph of $y = f(x)$ translated vertically, shifted up c units (or down if c is negative).
- The graph of $y = f(x - b)$ is the graph of $y = f(x)$ translated horizontally, shifted right b units (or left if b is negative).

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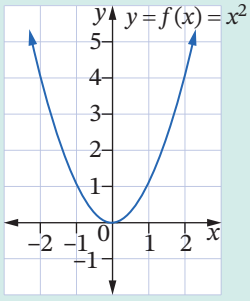
Example 6

If $y = f(x) = x^2$, then graph:

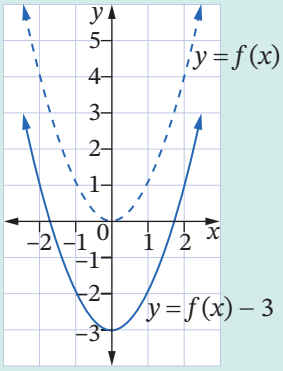
a $y = f(x)$ **b** $y = f(x) - 3$ **c** $y = f(x + 5)$

Solution

a

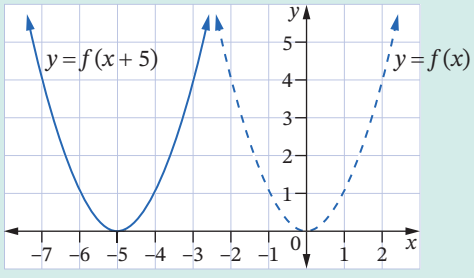


b



For $y = f(x) - 3$, the graph of $y = f(x)$ is shifted down 3 units.

c



For $y = f(x + 5)$, the graph of $y = f(x)$ is shifted 5 units to the left.

EXERCISE 13.03 ANSWERS ON P. 681

Graphing translations of functions U F R

1 If $y = f(x) = 2x$, then graph: R

a $y = f(x)$ **b** $y = f(x) + 1$ **c** $y = f(x) - 3$

d $y = f(x - 2)$ **e** $y = f(x + 4)$

2 If $y = f(x) = x^2$, then graph: R

a $y = f(x) - 3$ **b** $y = f(x) + 2$ **c** $y = f(x) - 1$

d $y = f(x - 3)$ **e** $y = f(x + 1)$ **f** $y = f(x + 2) - 1$

3 If $y = f(x) = x^3$, then graph: R

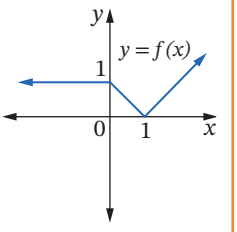
a $y = f(x - 1)$ **b** $y = f(x) + 2$ **c** $y = f(x + 2) + 1$

EXAMPLE 6

Foundation Standard Complex

PATH

4 The graph of $y = f(x)$ is shown. Sketch the graph of: R

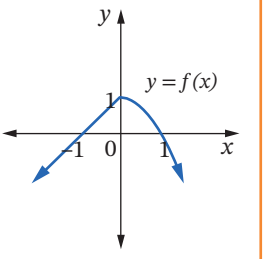


a $y = f(x) - 2$ **b** $y = f(x - 3)$

c $y = f(x + 2)$ **d** $y = f(x) + 3$

e $y = f(x + 1) - 3$

5 The graph of $y = f(x)$ is shown. Sketch the graph of: R



a $y = f(x) + 2$ **b** $y = f(x) - 3$

c $y = f(x - 1)$ **d** $y = f(x - 2) + 1$

13.04 Linear inequalities on the number plane

PATH

Worksheet
A page of lines and regions

The graph of a linear equation such as $y = 2x - 1$ on the number plane is a **line.**

The graph of a linear inequality such as $y > 2x - 1$ on the number plane is a **region.**

The line $y = 2x - 1$ divides the number plane into 2 regions or half-planes, one representing $y > 2x - 1$ and the other representing $y < 2x - 1$.

For all points within a region, the inequality holds true.

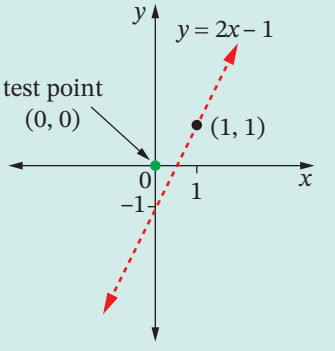
Example 7

Graph $y > 2x - 1$ on the number plane.

Solution

First, sketch the line $y = 2x - 1$ on the number plane, but sketch it as a broken line because it will NOT be included in the region.

The graph of $y > 2x - 1$ is the region **above** the line, but test a convenient point above the line such as (0, 0).



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Testing (0, 0) on $y > 2x - 1$:

LHS $= y$

$= 0$

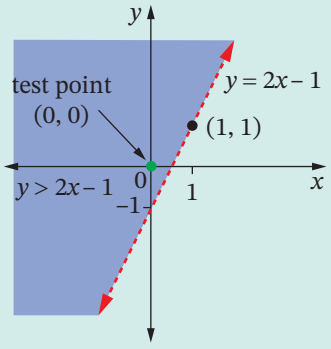
RHS $= 2x - 1$

$= 2 \times 0 - 1$

$= -1$

LHS > RHS, so (0, 0) is in the region.

Shade and label the region.



If when testing a point and the inequality is **false**, then shade the **other** half-plane region.

Example 8

Graph each region on a number plane.

a $y \leq x + 1$ **b** $x < 2$ **c** $y \geq -3$

Solution

a For the graph of $y \leq x + 1$, the line $y = x + 1$ is included in the region because '≤' includes 'equal to'.

Sketch $y = x + 1$ on the number plane as an unbroken line.

The graph of $y \leq x + 1$ includes the region **below** the line, but test a convenient point in the region such as (0, 0).

Testing (0, 0) on $y \leq x + 1$:

LHS $= y$

$= 0$

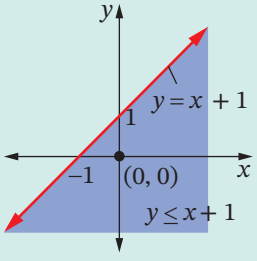
RHS $= x + 1$

$= 0 + 1$

$= 1$

LHS ≤ RHS, so (0, 0) is in the region.

Shade and label the region.

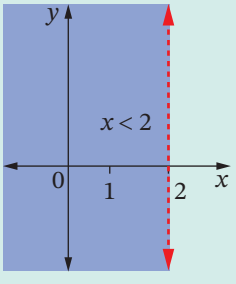


b For the graph of $x < 2$, the line $x = 2$ is not included in the region because '<' excludes 'equal to'.

Sketch $x = 2$ on the number plane as a broken vertical line.

The graph of $x < 2$ is the region **left** of the line, because all points in this region have x values less than 2.

Shade and label the region.



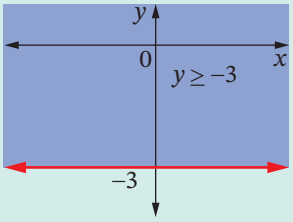
PATH

c For the graph of $y \geq -3$, the line $y = -3$ is included in the region because '$\geq$' includes 'equal to'.

Sketch $y = -3$ on the number plane as an unbroken horizontal line.

The graph of $y \geq -3$ includes the region **above** the line, because all points in this region have y values greater than -3.

Shade and label the region.



Graphing linear inequalities on the number plane

- Graph the linear equation as a broken line if the inequality involves $>$ or $<$.
- Graph the linear equation as an unbroken line if the inequality involves $\geq$ or $\leq$.
- Inequalities involving $>$ or $\geq$ are usually above the line.
- Inequalities involving $<$ or $\leq$ are usually below the line.
- Substitute a convenient point such as (0, 0) into the inequality.
- Shade and label the required region.

EXERCISE 13.04 ANSWERS P. 682

Linear inequalities on the number plane U F R C

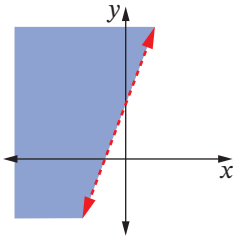
EXAMPLE 7

1 Graph each linear inequality on a number plane. R C

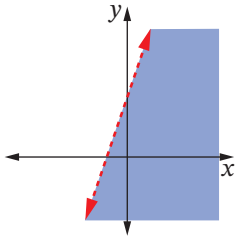
a	$y \geq 2x$	**b**	$y > 7 - x$	**c**	$y < 1 - x$
d	$y \geq 4x + 2$	**e**	$y > 2 - x$	**f**	$y \leq 3x - 4$
g	$3x + y > 6$	**h**	$x - 3y \leq 6$	**i**	$y + 2x - 4 \leq 0$

2 Which graph could be the region that represents $2x + y < 12$?
Select the correct answer **A**, **B**, **C** or **D**. R

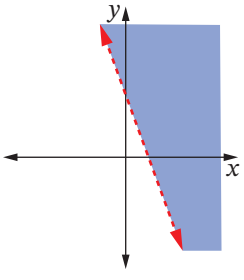
A



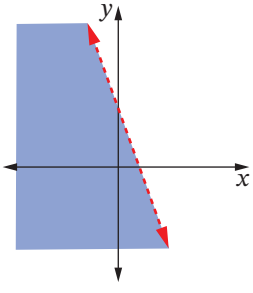
B



C



D



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PATH

EXAMPLE 8

13.04

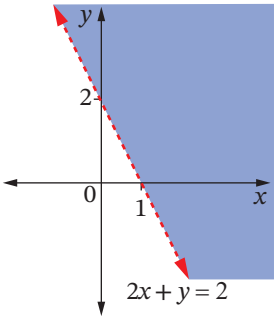
3 Graph each linear inequality on a number plane. R C

a $y > 2$ **b** $y \leq 1$ **c** $x \leq 3$

d $x \geq -4$ **e** $y < -2$ **f** $-1 \leq x \leq 4$

g $y \geq 5$ or $y \leq 1$ **h** $x \geq 0$ or $y \leq 2$ **i** $x < -2$ and $y > 4$

4 Which inequality is represented by this region on the number plane? Select **A**, **B**, **C** or **D**. R C



A $2x + y < 2$ **B** $2x + y > 2$

C $2x + y \leq 2$ **D** $2x + y \geq 2$

5 Which point does *not* lie in the region represented by the inequality $2y - x \geq 6$? Select **A**, **B**, **C** or **D**. R

A $(4, 0)$ **B** $(-6, 0)$ **C** $(0, 4)$ **D** $(-2, 8)$

6 Which region contains the point $(-6, -2)$? Select **A**, **B**, **C** or **D**. R

A $x < -8$ **B** $y < -3$ **C** $y + 2 \leq -2x$ **D** $y \leq x$

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